F1 RACE INTELLIGENCE

Project Design Phase-II

Requirement Analysis Report

Solution Requirements | Technology Stack | Data Flow Diagram & User Stories | Customer Journey Map

Prepared by

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Tool: Tableau | Dataset: Formula 1 Racing dataset (17 seasons, 6,464 race records)

28 October 2026

Project Background and Topic Information

Everything relevant about the topic, in one place

1. About the Project

F1 Race Intelligence is a Business Intelligence project built in **Tableau** by **Vijeta Nagda**. It converts historical Formula 1 race data into an interactive dashboard and a guided 9-scene story, so that people who work with F1 data can find answers about drivers, teams, circuits, tyres and championships in seconds instead of searching through spreadsheets.

The solution is organised around three analytical scenarios, 9 visualisations, one dashboard (fixed size 1400 × 1000, white theme, KPI row and filters) and one Tableau story.

2. About the Domain: Formula 1

- **What it is:** Formula 1 (F1) is the highest class of international single-seater motor racing, governed by the FIA (Fédération Internationale de l'Automobile). The FIA Formula One World Championship has run since 1950.
- **Season structure:** each season is a series of races called Grands Prix, held at circuits in many countries. A race weekend normally includes practice, qualifying and the race.
- **Two championships:** the Drivers' Championship (individual drivers) and the Constructors' Championship (teams, also called constructors). Points are awarded for finishing positions, and the totals decide the champions.
- **Points:** the finishing order of a race earns points for the top finishers (the current system gives 25 points to the winner, then 18, 15, 12, 10, 8, 6, 4, 2 and 1 down to tenth place). Points rules have changed over the decades, so this project uses the points recorded in the dataset.
- **Podium:** finishing in the top three positions of a race.
- **DNF (Did Not Finish):** a driver who does not complete the race, for example because of an accident, engine or gearbox failure, or another technical problem. DNF patterns show reliability.
- **Tyres:** teams choose between dry-weather compounds (soft, medium and hard) and wet-weather tyres (intermediate and full wet). Tyre choice and pit-stop strategy strongly influence race results.
- **Why data matters:** teams, broadcasters and regulators all use race statistics for strategy, storytelling and rule-making, which makes F1 a rich subject for analytics.

3. Problem Statement

F1 data is spread across many tables (race results, championship standings, circuits and related sheets) and is hard to read in raw form. Stakeholders cannot quickly see which drivers are reliable, which teams dominate, how championships were decided, which tyre compounds are used most, or which countries take part most often.

4. Objectives

- Bring the F1 dataset into Tableau and relate the tables into one analysable data model.
- Clean and prepare the data, and create calculated fields for the KPIs.
- Build 9 clear visualisations that answer real questions from three user scenarios.
- Combine the visuals into one interactive white-theme dashboard with a KPI row and filters (DNF Reason, Year of Race Date, Top N).
- Present the findings as a 9-scene Tableau story for easy walk-through and presentation.
- Document requirements, data flow, technology stack and the customer journey for the project.

5. Dataset

File	F1_Racing_Project_Dataset.xlsx (Microsoft Excel)
Size	6,464 race records covering 17 seasons
Main sheets	race_results, championship_standings, circuits and other related sheets
Typical fields used	Driver, team (constructor), circuit and country, race date (year), finishing position, points, DNF reason / status, tyre compound (exact column names follow the workbook)
Used for	Wins, podiums, points, DNF rate, season-by-season championships, tyre usage and country participation

6. Analytical Scenarios

Scenario / User	Goal	Key questions	Main visuals
Team Performance Strategist	Judge and compare how teams and drivers perform across seasons	Which teams and drivers score the most points, wins and podiums? How did championships develop?	Team ranking, team championship, driver leaderboard, points trajectory
Broadcast Analyst	Find quick, accurate storylines for commentary and TV graphics	Where are the drivers from? Who stands on the podium most? Which countries take part most?	Map, podium bubbles, country participation treemap
FIA Regulation Analyst	Review reliability and tyre usage to support safety and rule decisions	Which drivers and teams have most DNFs and why? How are tyre compounds used?	Driver reliability, tyre compound pie

7. The 9 Visualisations

#	Visualisation	Purpose
1	Map (geographic driver distribution)	Shows where drivers and races come from around the world.
2	Driver reliability	Shows DNF counts / rates by driver and DNF reason.
3	Driver leaderboard	Ranks drivers by points, wins or podiums (Top N).
4	Team ranking	Ranks teams (constructors) by performance.
5	Team championship	Shows season-by-season championship standings.
6	Tyre compound pie	Shows the share of each tyre compound used.
7	Country participation treemap	Shows which countries take part most, by area size.
8	Podium bubbles	Shows podium achievements as bubbles sized by count.
9	Points trajectory	Shows how points build up over time for selected drivers or teams.

8. Dashboard and Story Design

- **Dashboard:** white theme, fixed size 1400 × 1000, title banner, KPI row and the visuals arranged in a clean grid.
- **Filters:** DNF Reason, Year of Race Date and a Top N parameter that apply across the relevant visuals.

- **Story (9 scenes):** KPIs overview, geographic driver distribution, driver reliability / DNF, driver leaderboard, team ranking, season-by-season championship, tyre strategy, country-level race participation, podium achievements.

9. Key Terms (Glossary)

Term	Meaning
BI	Business Intelligence – turning data into dashboards and insights for decisions.
KPI	Key Performance Indicator, such as total wins, podiums, points or DNF rate.
DNF	Did Not Finish – a driver who does not complete the race.
Constructor	An F1 team that builds and races its own car; competes for the Constructors' Championship.
Podium	Top three finishing positions in a race.
Compound	The type of tyre rubber (soft, medium, hard, intermediate, wet).
Top N	A parameter that limits a chart to the best N items (for example the top 10 drivers).
Calculated field	A formula created in Tableau to derive a new measure or dimension.
Story	A Tableau sequence of scenes (sheets or dashboards) that tells a data narrative.
DFD	Data Flow Diagram – shows how data moves through a system.

10. Expected Benefits

- Faster, evidence-based decisions for strategists, broadcasters and regulators.
- One shared view of 17 seasons of F1 data instead of many separate tables.
- Interactive exploration with filters, tooltips and Top N controls, no coding needed.
- A repeatable workbook that can be refreshed when new seasons are added.

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	28 October 2026
Team ID	EN25CS3011046
Project Name	F1 Race Intelligence (by Vijeta Nagda)
Maximum Marks	4 Marks

Functional Requirements

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority (High/Medium/Low)
1	Data Access & Integration	• Connect Tableau to F1_Racing_Project_Dataset.xlsx using the Excel connector. • Access the race_results, championship_standings, circuits and related sheets (6,464 race records, 17 seasons). • Relate the tables on common keys (race, driver, team, circuit) to form one data model.	High
2	Data Processing	• Clean missing values, duplicates and inconsistent names. • Standardise DNF reason / status values for the DNF filter. • Extract the year from the race date for the Year filter. • Create calculated fields: win flag, podium flag, DNF flag, points totals.	High
3	Business Analysis	• Calculate KPIs: total races, wins, podiums, total points, DNF rate and average points per race. • Rank drivers and teams using a Top N parameter. • Compute season-by-season championship standings and points trajectory. • Aggregate tyre compound share and country participation.	High
4	Business Insights & Analytical Outputs	• Provide 9 visualisations: map, driver reliability, driver leaderboard, team ranking, team championship, tyre compound pie, country participation treemap, podium bubbles, points trajectory. • Provide one white-theme dashboard (1400 × 1000) with title banner and KPI row. • Provide a 9-scene Tableau story.	High
5	User Interaction & Analysis	• Filter by DNF Reason, Year of Race Date and Top N. • Hover tooltips for exact values; click to highlight related marks. • Compare teams and drivers; explore from season level to driver or team level.	High
6	Data Update & Reporting	• Refresh the data source when the Excel file is updated. • Export the dashboard and story as PDF or image for reports. • Optionally publish the workbook to Tableau Public to share a link.	Medium

Non-functional Requirements

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The dashboard must be easy to read and use without training: clear titles, consistent white theme, labelled filters, readable fonts and a logical layout with a KPI row on top.
NFR-2	Security	The dataset is historical, non-personal public-type data. The workbook is stored on a protected device or drive; when published, only the intended audience gets the link, and no confidential data is included.
NFR-3	Reliability	KPIs and charts must match the source data exactly. Calculated fields are tested against the Excel file, and the workbook is saved with backup copies.
NFR-4	Performance	Dashboard filters and parameters should respond within a few seconds on a normal laptop; using extracts and limiting marks (Top N) keeps the dashboard fast.
NFR-5	Availability	The workbook is available whenever Tableau Desktop is opened locally; the published Tableau Public version (if used) is available online at any time.
NFR-6	Scalability	New seasons or additional tables can be added to the Excel source and refreshed without redesigning the dashboard; the data model can later move to a database or data warehouse.

Requirement Traceability

Functional requirement	User stories covered	Where it appears
FR-1 Data Access	USN-15	Data Source page in Tableau
FR-2 Data Processing	USN-16	Calculated fields, cleaned data
FR-3 Business Analysis	USN-1 to USN-4, USN-9	KPI row, ranking and trend visuals
FR-4 Insights & Outputs	USN-5 to USN-8, USN-10, USN-18	9 visuals, dashboard, story
FR-5 User Interaction	USN-11 to USN-13	Filters, Top N parameter, tooltips
FR-6 Update & Reporting	USN-14, USN-17	Refresh, PDF / image export

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	28 October 2026
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Project Name	F1 Race Intelligence (by Vijeta Nagda)
Maximum Marks	2 Marks

Define Your Technology Stack

The technologies, platforms, data sources and BI tools used to build the F1 Race Intelligence solution are listed below.

S.No	Component	Description	Technology	Justification / Purpose
1.	Programming / Query Language	Language used to query, filter, transform, aggregate and analyse data.	Tableau calculation language (calculated fields, LOD expressions, table calculations); Excel formulas for checking	Lets us derive wins, podiums, DNF flags and rankings inside Tableau without separate code.
2.	Database / Data Warehouse	Used to store and manage structured business data for analysis and reporting.	Microsoft Excel (.xlsx) – F1_Racing_Project_Dataset.xlsx	Simple, familiar and sufficient for 6,464 records; easy to update with new seasons.
3.	Data Ingestion	Used to load data from source files into the analytical environment.	Tableau Data Connections (Excel connector) with relationships	Direct connection to the workbook; relationships keep the related tables together without manual joins.
4.	Data Transformation and Business Logic	Tools / logic used to clean, join, filter, aggregate data and calculate KPIs.	Tableau Desktop (Data Source page, calculated fields, filters, parameters)	One tool for cleaning and KPI logic, which keeps the project simple and traceable.
5.	Data Visualization / Analytics	Used to create interactive dashboards, charts, reports, KPIs and visual analysis.	Tableau Desktop – worksheets, dashboard, story	Provides the map, treemap, pie, bubble and trend charts needed for the 9 visuals and the 9-scene story.
6.	Front-End / User Interface	Interface through which users interact with dashboards, filters, parameters and visual insights.	Tableau Desktop / Tableau Public dashboard	Users filter and explore directly in the dashboard; Tableau Public allows sharing by link.
7.	Reporting / Export	Used to share results outside Tableau.	PDF and image export; Microsoft Word / PDF for documentation	Allows results to be included in project reports and presentations.

Project Design Phase-II

Data Flow Diagram & User Stories

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Maximum Marks	4 Marks

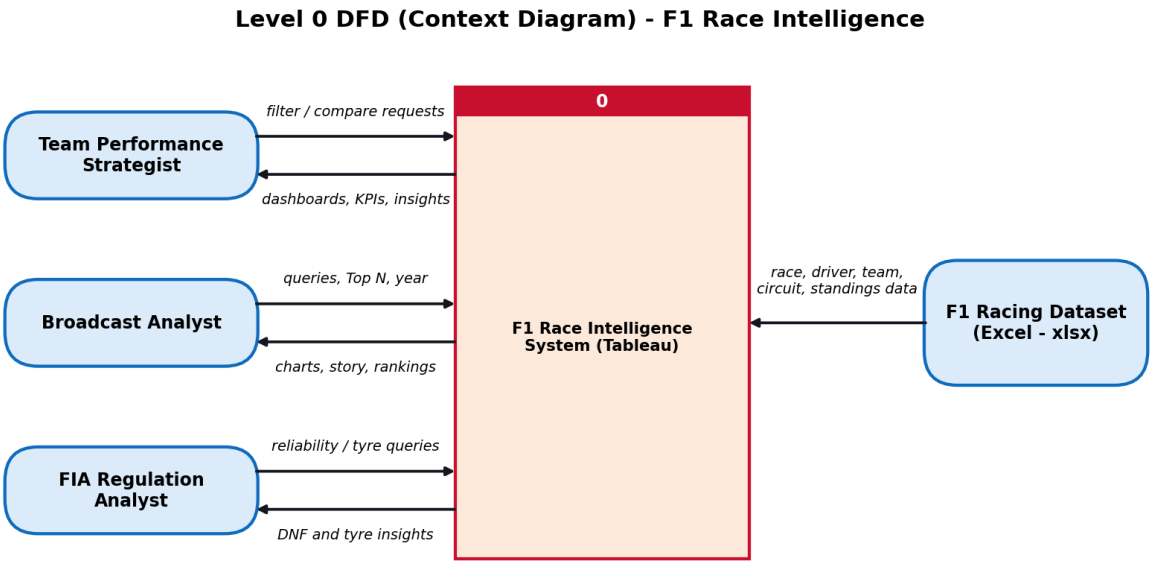
Data Flow Diagrams

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. It shows how data enters and leaves the system, what changes the information, and where data is stored. The DFDs below describe how data moves through the F1 Race Intelligence solution.

DFD Symbol Legend

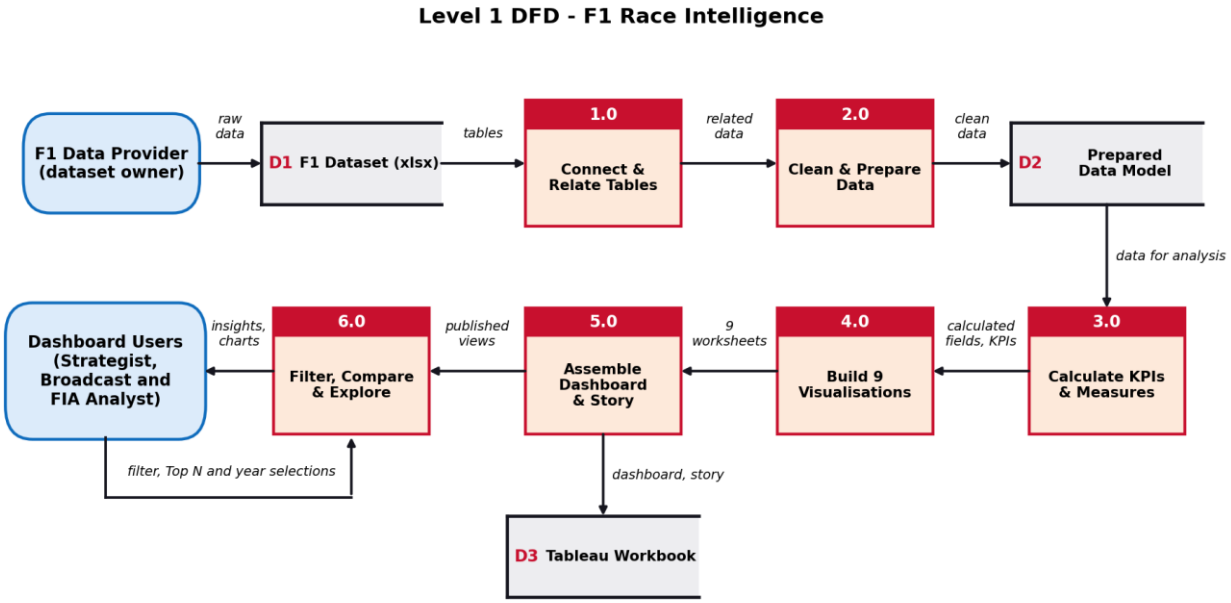
Symbol	Name	Description
Oval / Rounded shape	External Entity	A person, organisation or system outside the project that sends or receives data (e.g. Dashboard User, F1 Data Provider).
Rectangle with numbered header	Process	An activity that transforms incoming data into outgoing data (e.g. Clean & Prepare Data).
Open rectangle (solid fill, no number on the shape body)	Data Store	A place where data is held for later use (e.g. F1 Dataset, Prepared Data Model).
Labelled arrow	Data Flow	The movement of data between entities, processes and data stores, labelled with the data being passed.

Level 0 DFD (Context Diagram)



The context diagram shows the whole system as a single process. The three user groups send filter and query requests and receive dashboards, KPIs, charts and the story. The F1 dataset is the only data input.

Level 1 DFD



ID	Process	Input	Output	Data store
1.0	Connect & Relate Tables	Raw sheets from the F1 dataset	Related tables	D1 F1 Dataset (xlsx)
2.0	Clean & Prepare Data	Related tables	Clean data (standardised DNF reasons, year field)	D2 Prepared Data Model
3.0	Calculate KPIs & Measures	Clean data	Calculated fields, KPIs, rankings	D2 Prepared Data Model
4.0	Build 9 Visualisations	KPIs and measures	9 worksheets	D3 Tableau Workbook
5.0	Assemble Dashboard & Story	9 worksheets	Dashboard (1400 × 1000) and 9-scene story	D3 Tableau Workbook
6.0	Filter, Compare & Explore	User filter, Top N and year selections	Updated charts and insights for users	D3 Tableau Workbook

User Stories

The table lists the user stories for the F1 Race Intelligence product. Priorities and releases (sprints) are the proposed delivery plan for the project.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Team Performance Strategist	Dashboard Overview	USN-1	As a strategist, I can see a KPI row (races, wins, podiums, points) at the top of the dashboard so that I can judge performance at a glance.	KPI values are shown and match totals calculated from the dataset.	High	Sprint-1
	Team Analysis	USN-2	As a strategist, I can view teams ranked by points and wins so that I can compare team performance.	Team ranking chart is sorted correctly and changes when Top N is changed.	High	Sprint-1
	Championship Analysis	USN-3	As a strategist, I can view season-by-season championship standings and the points trajectory so that I can see how titles were won.	Team championship and points trajectory charts show the selected seasons correctly.	High	Sprint-2
	Driver Analysis	USN-4	As a strategist, I can view a driver leaderboard so that I can identify the top performers.	Leaderboard lists the Top N drivers by points / wins / podiums.	High	Sprint-1
Broadcast Analyst	Geographic Analysis	USN-5	As a broadcast analyst, I can see a world map of driver and race locations so that I can find geographic storylines.	Map shows locations and tooltips with driver / race details.	Medium	Sprint-1
	Podium Analysis	USN-6	As a broadcast analyst, I can see podium achievements as bubbles so that I can quickly spot the most successful drivers.	Bubble size reflects podium count and tooltips show exact numbers.	Medium	Sprint-2
	Participation Analysis	USN-7	As a broadcast analyst, I can view a treemap of country participation so that I can show which nations are most involved.	Treemap areas are proportional to participation counts.	Medium	Sprint-2
	Storytelling	USN-8	As a broadcast analyst, I can walk through a 9-scene story so that I can present the findings in order.	All 9 scenes open in order with a caption on each scene.	High	Sprint-3
FIA Regulation Analyst	Reliability Analysis	USN-9	As an FIA analyst, I can view DNF counts by driver and team and filter by DNF reason so that I can assess reliability.	DNF chart updates when the DNF Reason filter is changed.	High	Sprint-1
	Tyre Analysis	USN-10	As an FIA analyst, I can view the share of each tyre compound used so that I can review tyre usage.	Pie chart shows each compound with its percentage.	Medium	Sprint-2

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
General Viewer (F1 fan / student)	Filtering	USN-11	As a viewer, I can filter all visuals by Year of Race Date so that I can focus on one season or period.	Selecting a year updates every connected visual.	High	Sprint-2
	Filtering	USN-12	As a viewer, I can hover over marks to see tooltips so that I can read exact values.	Tooltip appears with correct values on every chart.	Low	Sprint-3
	Filtering	USN-13	As a viewer, I can change the Top N parameter so that I can choose how many drivers or teams are shown.	Charts show exactly N items after the change.	Medium	Sprint-2
	Reporting	USN-14	As a viewer, I can export the dashboard as a PDF or image so that I can use it in reports.	Export file shows the full dashboard clearly.	Low	Sprint-3
Dashboard Developer / Administrator (Vijeta Nagda)	Data Access & Integration	USN-15	As the developer, I can connect the Excel dataset and relate the tables so that all data is available in one model.	All sheets are connected and record counts match the source (6,464 race records).	High	Sprint-1
	Data Processing	USN-16	As the developer, I can clean the data and create calculated fields so that KPIs are accurate.	Calculated fields return values that match manual checks.	High	Sprint-1
	Data Update	USN-17	As the developer, I can refresh the data source when the Excel file changes so that the dashboard stays up to date.	After refresh, new records appear in all visuals.	Medium	Sprint-3
	Dashboard Design	USN-18	As the developer, I can apply the white theme, title banner and fixed 1400 × 1000 layout so that the dashboard looks consistent and professional.	Dashboard shows consistent layout without overlap or scrolling.	Medium	Sprint-2

Project Design Phase-II

Customer Journey Map

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Maximum Marks	2 Marks

The map shows the experience of a typical dashboard user (strategist, broadcast analyst or FIA analyst) stage by stage.

Phase of Journey	Stage 1: Awareness / Need	Stage 2: Access	Stage 3: Exploration	Stage 4: Insight & Decision	Stage 5: Share & Follow-up
Actions – What does the customer do?	• Receives a question about drivers, teams or reliability. • Looks for F1 data to answer it. • Learns about the F1 Race Intelligence dashboard.	• Opens the Tableau dashboard or workbook. • Reads the title and KPI row. • Checks the filters available.	• Applies Year, DNF Reason and Top N filters. • Compares teams and drivers. • Hovers over charts for details.	• Identifies the key finding. • Checks it in the story scenes. • Decides on strategy, commentary or rule input.	• Exports the dashboard as PDF or image. • Shares the link or report. • Returns for the next race or season.
Touchpoint – What part of the service do they interact with?	• Meeting, email or news item. • Project documentation.	• Tableau Desktop / Tableau Public. • Dashboard title banner and KPI row.	• Filters, parameters and tooltips. • The 9 visualisations.	• 9-scene story. • Dashboard summary views.	• PDF / image export. • Tableau Public link.
Customer Thought – What is the customer thinking?	• “I need facts quickly.” • “Where can I find all this data?”	• “Is this easy to use?” • “Can I trust these numbers?”	• “Which team is most reliable?” • “Show me just this season.”	• “Now I can see the pattern.” • “This supports my decision.”	• “I can reuse this next season.” • “Others should see this too.”
Customer Feeling – What is the customer feeling?	Overloaded and unsure where to start.	Curious and a little cautious.	Engaged and in control.	Confident and satisfied.	Proud and motivated to reuse it.
Process Ownership – Who is in the lead on this?	Customer (analyst) with the project owner Vijeta Nagda.	Vijeta Nagda (dashboard developer).	Customer, supported by the dashboard design.	Customer / decision maker.	Vijeta Nagda for updates; customer for sharing.
Opportunities – How can we improve this stage?	• Add a short project introduction. • Share a quick-start guide.	• Add a help / how-to-use text on the dashboard. • Show data source and refresh date.	• Add filter actions between charts. • Add more tooltips and reset buttons.	• Add short insight captions on each story scene. • Highlight key outliers.	• Add one-click export instructions. • Add new seasons and a prediction view later.

Conclusion

This requirement analysis defines what the F1 Race Intelligence solution must do, which technologies it uses, how data flows through it, who uses it and how they experience it. The dataset, Tableau build, dashboard and story together turn 17 seasons of Formula 1 data into insights that help strategists, broadcasters and regulators. Prepared by **Vijeta Nagda**.